<u>**GUJARAT COUNCIL ON SCIENCE AND TECHNOLOGY**</u>
Dept. of Science and Technology, Govt. of Gujarat

ROBOFEST- GUJARAT 5.0

Guidelines for submission of Ideation Proposal
Senior category

All teams must be present and ready to perform their challenge. The committee may assign any team to the task at random. Any kind of absence and lack of readiness will be subject to suitable actions, including the deduction of score/disqualification.

Students are required to perform their task and answer the questions. Mentors should not be involved in the task. The committee may allow mentors during question-answer if it feels necessary, depending on the situation.

Sample images given in these guidelines are only for visual understanding

<u>Aerial Robotics: Minefield Navigation Challenge</u>

Game Description:

In this senior-level challenge, teams will design and deploy a swarm of autonomous aerial vehicles (comprising a minimum of four micro drones) to help a virtual "person-at-risk" safely cross a simulated 100-meter-long minefield within 10 minutes. The mission demands advanced aerial coordination, obstacle avoidance, swarm communication, and safe path mapping without external control systems. This challenge promotes real-world relevance in defense, disaster response, and humanitarian assistance.

Challenge Objective:

- Safely navigate a person-at-risk from pre-defined Start Zone to Exit Zone across a 100-meter "minefield".
- Each team must launch and operate a minimum of 4 autonomous micro aerial vehicles (under 500 gram/0.5 Kg each).
- Only onboard processing is allowed. No external data links (except kill-switch/safety override).
- The drones must:
  - Map surface-level "mines" (visibly marked for simulation).
  - Avoid redundancy in mine detection.
  - Generate and communicate a 1-meter clearance safe path.
  - Mark the safe path visually (LED, coloured powder, or markers).
  - Track the human and re-route if the path changes.

Challenge Tasks:

1. Take-off and Human Gesture Recognition
   - Drones activate on hand/voice command of the "person-at-risk".
   - Must recognize simple gestures (forward, pause, scan left/right).
2. Mine Detection and Mapping
   - Use onboard optical/IR sensors to identify mine locations.
   - Create a digital map stored in local memory for coordination.
3. Swarm Coordination
   - Coordinate with other drones to avoid rechecking the same area.
   - Each drone scouts a separate section of the field.
4. Safe Path Marking
   - Highlight safe path using visual cues (LEDs or coloured chalk/powder).
   - Maintain a 1-meter safety buffer from detected mines.
5. Path Communication and Human Tracking
   - Communicate the path to "person-at-risk" visually (arrow pattern or corridor lights).

- Dynamically update the path if the human deviates or the minefield changes.

6. Mission Completion
    - Person-at-risk crosses safely under drone guidance within 10 minutes.
    - The drone swarm must land autonomously once the mission ends.

Game Field Design:

- Field Size: 20m width × 100m length.
- Start Zone (2m × 20m): Drone deployment and human starting area.
- Minefield Zone (90m × 20m): Contains 20–40 randomly placed "mines" (plastic disks or markers).
- Exit Zone (8m × 20m): Finish line for the person-at-risk.
- Marked Zones:
    - Mines simulated using flat discs (20–30 cm radius).
    - Drones must maintain 1m exclusion zone around each mine.
    - Obstacles like poles (1.5m height) may simulate trees/walls.

Scoring Criteria:

Task	Points
Safe takeoff and swarm activation	10 points
Gesture/voice command recognition	10 points
Mine detection & correct mapping	30 points
Safe path creation & marking	20 points
Person crosses safely	20 points
Time Bonus (per minute saved)	+5 points/min
Collision or unsafe proximity	-10 points
Drone crash or surface contact	-20 points